



Machine-Learning Assisted Materials Discovery of a Blue Emitter for a More Efficient and Durable OLED Device

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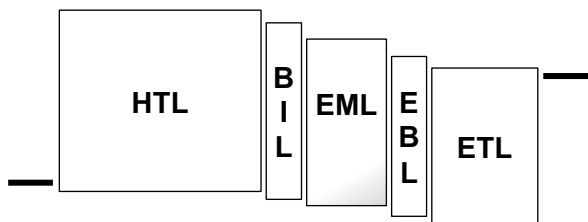
CAE Team

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Introduction

- **Phosphors** and **TADF** materials are of interest for a more **efficient** OLED device.
- **Chemical degradation** of an emitter is critical for a more **durable** OLED device.

External Quantum Efficiency (EQE, $\eta_{external}$)



$$\eta_{external} = \gamma \cdot \phi \cdot \eta_{exciton} \cdot \eta_{out}$$

γ : ratio of electron to hole (0.3 ~ 0.9)

(Device thickness: LED ~100 μm , OLED < 1 μm)

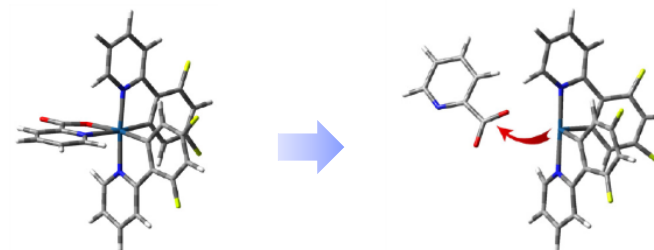
ϕ : internal quantum efficiency (based on molecule)

$\eta_{exciton}$: fraction of luminescent excitation

(Spin statistics: Fluorescence 25%, Phosphorescence 100%)

η_{out} : light out coupling efficiency

Stability of cyclometalated complexes



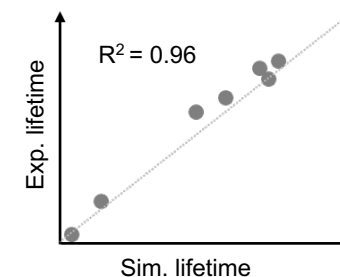
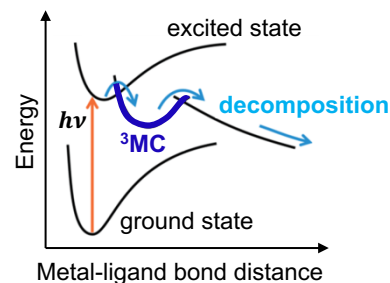
Flrpic

Metal-Ligand Decomposition

Chem. Mater. 31, 2269–2276 (2019)

Higher ^3MC energy level

$\Rightarrow$ Higher chance of longer lifetime



Introduction

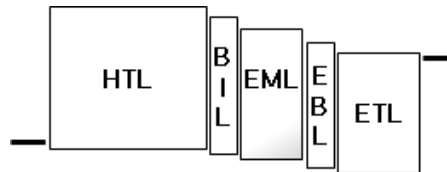
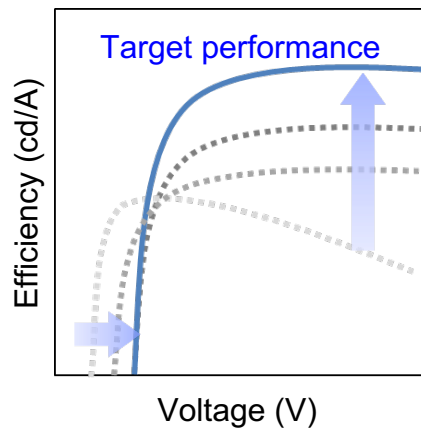
- We aim to **design a material that exhibits properties in the target** range of our interest.
- The target range of properties is induced from a combination of material properties of a device that could yield a better device performance.

Target efficiency

Layer properties

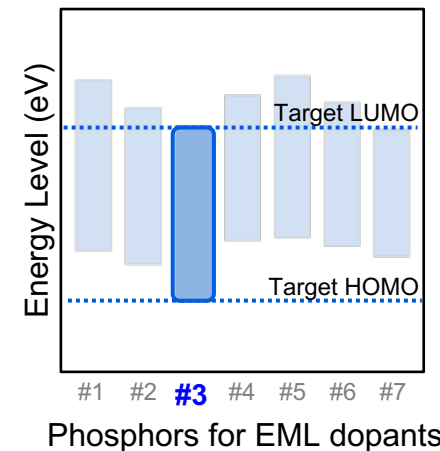
Materials

Best performance

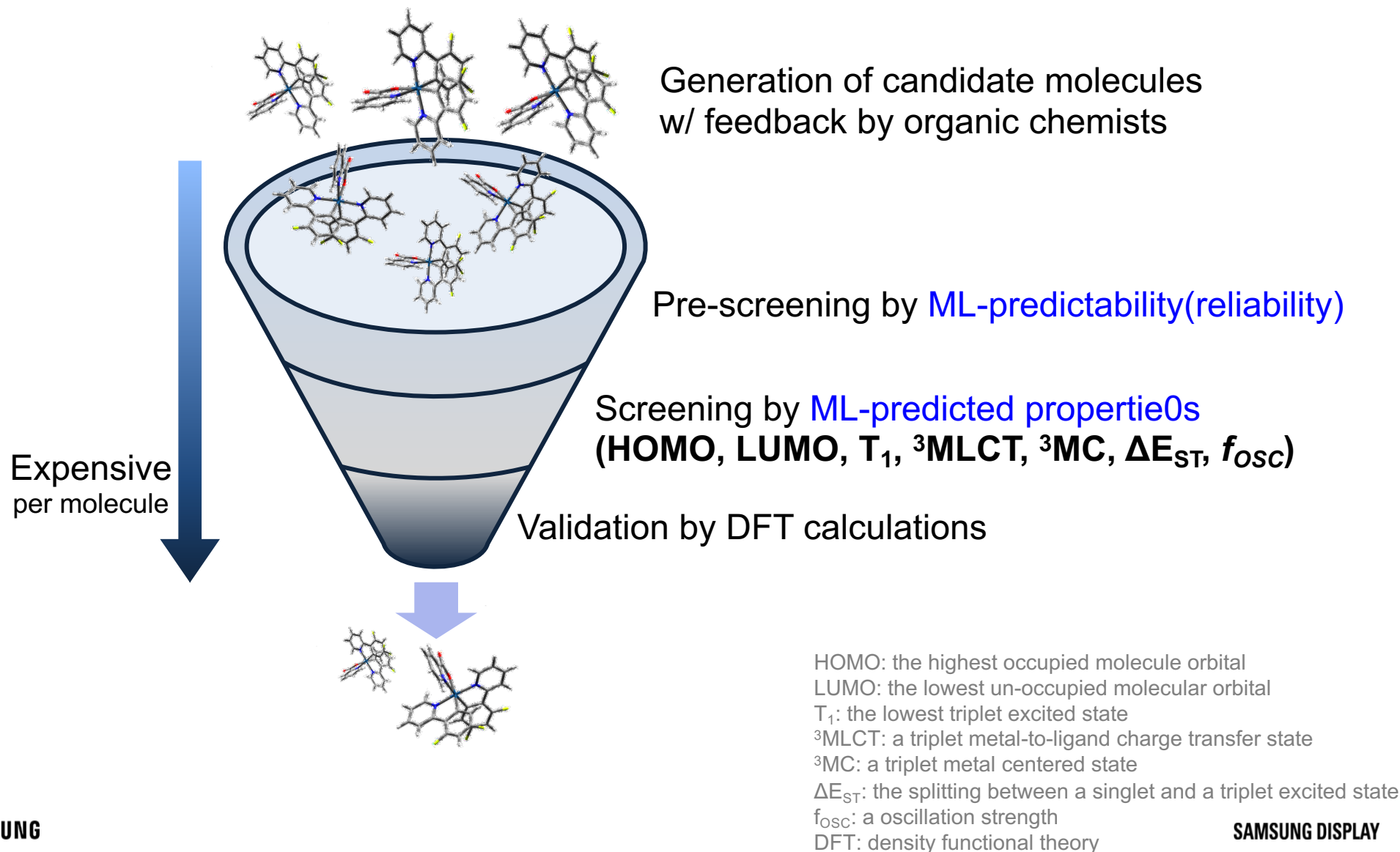


	HTL	BIL	EML	Buffer	ETL
Thickness	1300	50	200	50	350
Nc	1e+21	1e+21	1e+21	1e+21	1e+21
Nv	1e+21	1e+21	1e+21	1e+21	1e+21
LUMO	-1.72	-1.82	-1.82	-1.87	-2.1
HOMO	-5.3	-5.4	-5.45	-5.5	-5.6
Electron mobility	0.000002	1E-07	0.000002	0.0001	2e-5
Hole mobility	0.000000	0.000006	0.000008	0.0004	1e-5
Electron Charact. Field	0.01	0.001	0.01	0.001	1e-4
Hole Charact. Field	0.001	0.001	0.001	0.001	5e-3
Acceptor electron cap. ...	1e-20	1e-20	1e-20	1e-20	1e-20
Acceptor hole cap. Xsec	1e-20	1e-20	1e-20	1e-20	1e-20
Donor electron cap. Xsec	1e-20	1e-20	1e-20	1e-20	1e-20
Donor hole cap. Xsec	1e-20	1e-20	1e-20	1e-20	1e-20
Acceptor Trap density	1e+15	1e+17	1e+15	1e+17	2.1e+15
Donor Trap Density	1e+17	1e+17	1e+17	1e+17	7e+15
Acceptor Trap Charact. te	1500	1500	1500	1500	1500
Donor Trap Charact. te	1500	1500	1500	1500	1500
Conduction Band Gauss	0.2	0.2	0.2	0.2	0.1
Valance Band Gaussian	0.2	0.2	0.2	0.2	0.1
Permittivity	4	3.5	4	4	5
TAUS	1e-07	1e-07	1e-07	1e-07	1e-7
TAUT	1e-5	1e-5	1e-5	1e-5	1e-5
KnrS	0.0526e7	0.6667e7	0.0526e7	0e7	0
KnrT	0	0	0	0	0
Langvin to Singlet Frac.	0.25	0.25	0.25	0.25	0.1
Ktt to singlet Fraction	0.25	0.25	0.25	0.25	0.1
Singlet Diffusivity	1e-06	1e-06	1e-06	1e-06	1e-6
Triplet Diffusivity	5.32e-06	6.32e-06	1.03e-06	1.03e-06	0

Design of best materials



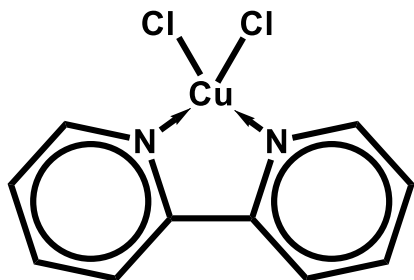
Introduction: virtual high-throughput screening (vHTS)



Building a database of candidate materials **structures** (1)

Canonical SMILES description of molecules by Daylight Chemical Information Systems

- Coordinate (dative) bonds of metal-ligand complexes require a careful attention in SMILES description to be used throughout various tools and python libraries.



SMILES description	
No description for dative bonds	<chem>Cl[Cu]1(Cl)n2cccc2c2cccn12</chem>
RDKit ^[1]	<chem>Cl[Cu]1(Cl)<-n2cccc2c2cccn->12</chem>
Obabel ^[2]	<chem>Cl[Cu]1(=n2cccc2c2n=1cccc2)Cl</chem>
Chemdraw/office	<chem>Cl[Cu-2]1(Cl)[n+]2cccc2c2[n+]1cccc2</chem>

- We have devised a rule that can be universally applied throughout many tools and libraries without an error.

Building a database of candidate materials **structures** (2)

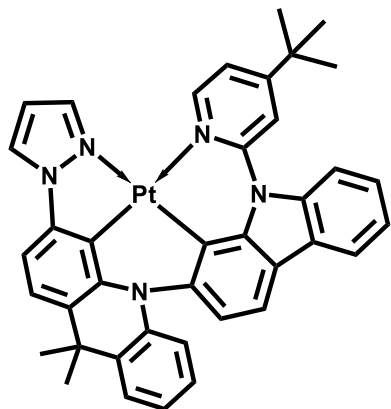
- Given feedback by organic chemists, we have built a molecule generation tool for metal-ligand complexes by enumerating every possible ligand fragments around a metal atom.

Target materials group

Enumeration

1.5M Candidate materials

w/ feedback by organic chemists



Appl. Phys. Lett. 110, 113301 (2017)



Cn1...(c1)[Pt]13c4n...ncn...c3c...3)ccn4
Cn1...(c1)[Pt]13c4n...ncc...c3c...3)ccc4
Cn1...(c1)[Pt]13c4n...ccn...c3c...3)cnn4
Cn1...(c1)[Pt]13c4n...ncn...c3c...3)ncn4
Cc1...(n1)[Pt]13c4n...ncn...c3c...3)ncn4
Cc1...(c1)[Pt]13c4n...ncc...c3c...3)ncn4
Cc1...(n1)[Pt]13c4n...ccn...c3c...3)ncn4
:
:



Building a database of material **properties** by ab-initio calculations

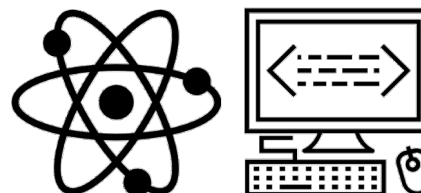
- For 1,000 molecules that are sampled from 1.5million candidate structures, we've built a property database by the automation of ab-initio calculations.

Sampled 1K molecules

Automated property calculations

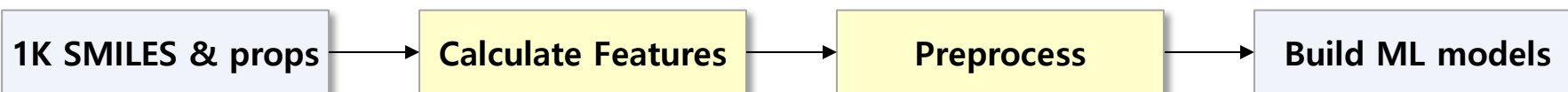
Ab-initio calculations based on density functional theory (DFT)

Cn1...(c1)[Pt]13c4n...ncn...c3c...3)ccn4
Cn1...(c1)[Pt]13c4n...ccn...c3c...3)cnn4
Cn1...(c1)[Pt]13c4n...ncc...c3c...3)ccc4
Cc1...(c1)[Pt]13c4n...ncn...c3c...3)ncn4
Cc1...(n1)[Pt]13c4n...ncn...c3c...3)ncn4
Cn1...(c1)[Pt]13c4n...ncn...c3c...3)ncn4
Cc1...(c1)[Pt]13c4n...ncc...c3c...3)ncn4
Cc1...(n1)[Pt]13c4n...ccn...c3c...3)ncn4
:
:

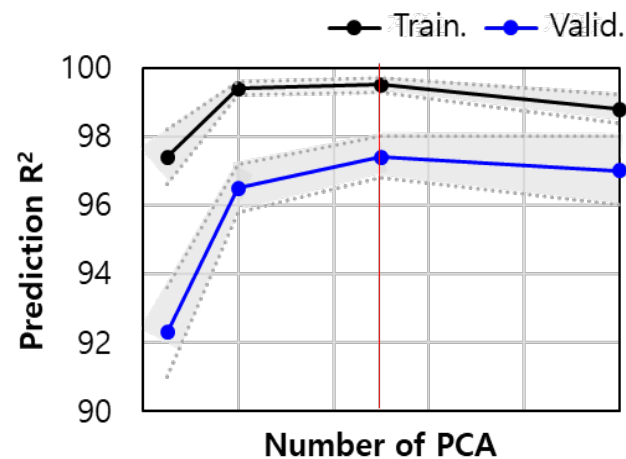
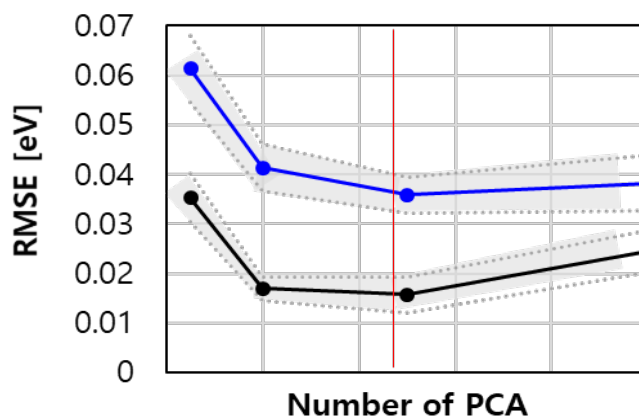
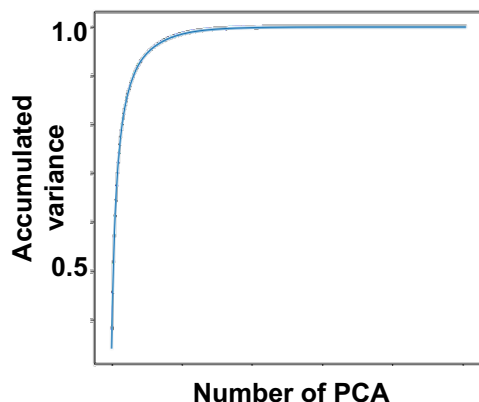


SMILES	³ MC	T ₁	³ MLCT	HOMO	LUMO
Cn1...(c1)[Pt]13c4n...ncn...c3c...3)ccn4	0.4	2.2	3	-5.5	-2.0
Cn1...(c1)[Pt]13c4n...ncc...c3c...3)ccc4	0.2	2.7	8	-5.2	-2.1
Cc1...(c1)[Pt]13c4n...ncn...c3c...3)ncn4	0.3	2.5	10	-5.3	-2.2
:	:	:	:	:	:

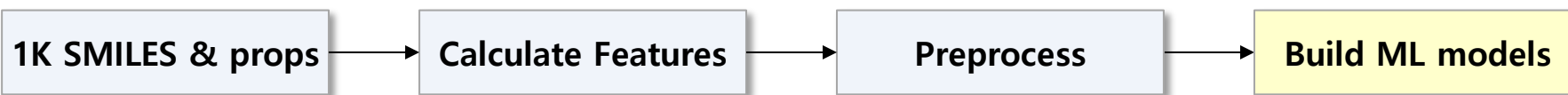
Machine learning models for property prediction (1)



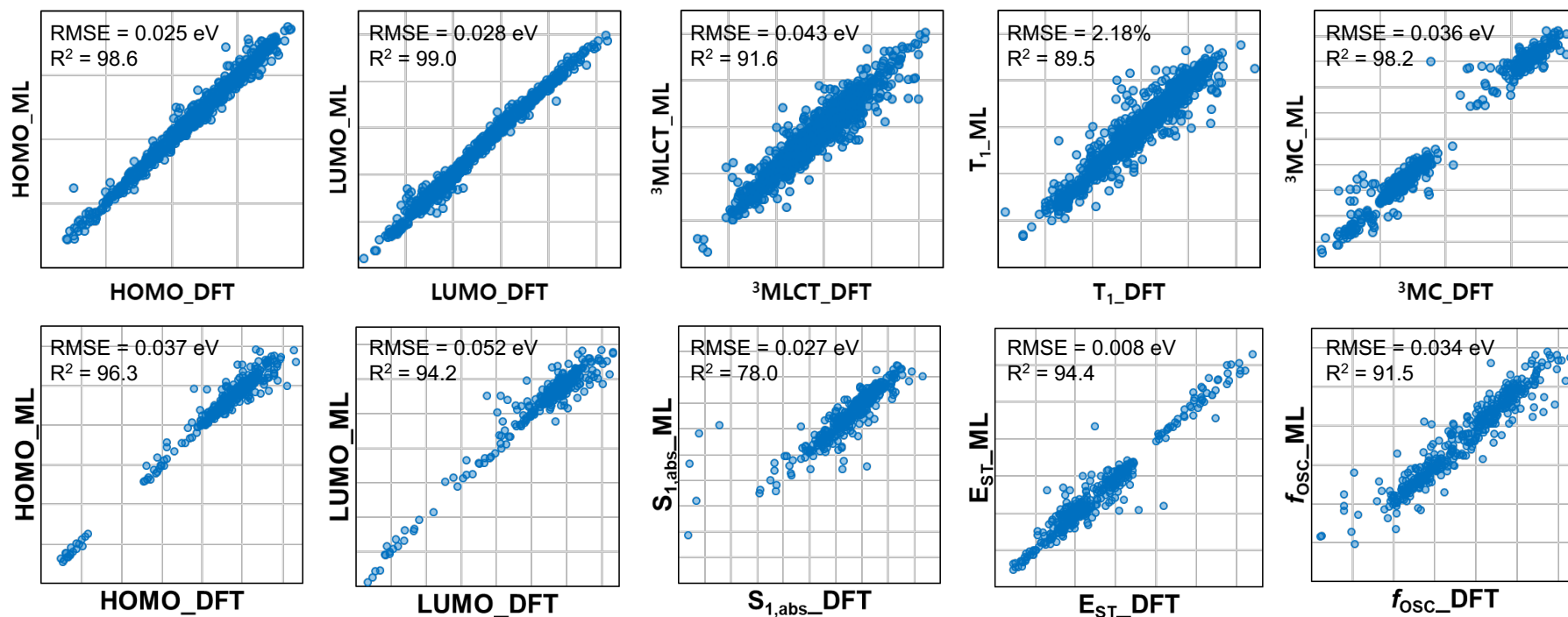
- Calculate descriptors based on a 2-dimensional structure.
- Remove outliers & nearly constant & highly correlated features.
- Reduce dimensions further by PCA, if appropriate.



Machine learning models for property prediction (2)



- **Machine learning models for DFT property prediction: An ensemble of 3-5 models**
 - Linear & Non-linear models (using Scikit-learn^[1])
 - Artificial neural networks (using keras^[2])
- **10-fold cross validation results**



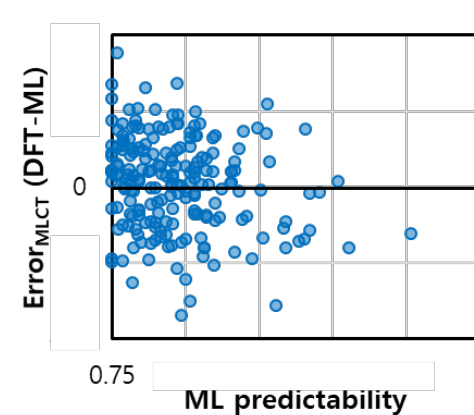
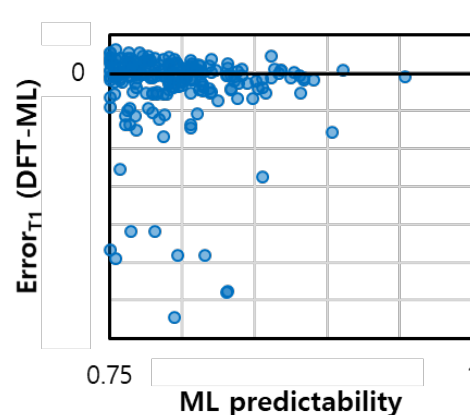
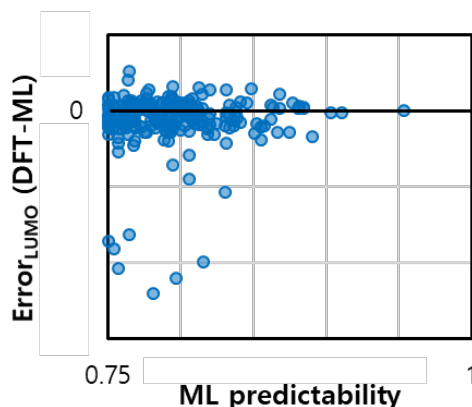
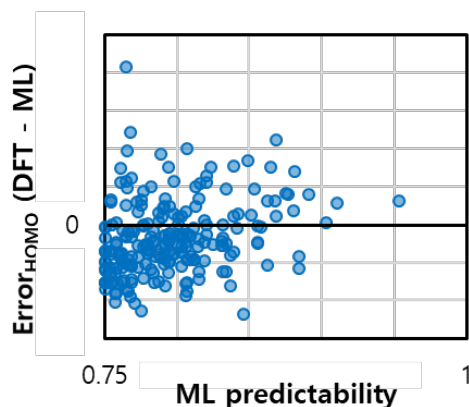
Inference of ML models & error estimations

- **ML-predictability (reliability) scale**
to determine whether a particular ML model is appropriate to be applied for the inference of a test molecule or not, based solely on structural information.
⇒ Higher ML-predictability
→ Higher chance of small errors
- We've implemented our ML-predictability scale to pre-screen the candidate molecules before the virtual HTS by ML property prediction models.

[RMSE for a test set]

60 molecules randomly sampled from 1.5M candidates

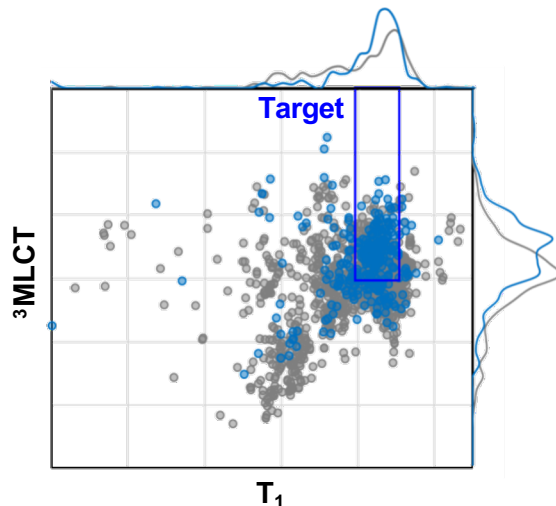
	Predictability scale	
	Higher	Lower
HOMO	0.06 eV	0.29 eV
LUMO	0.09 eV	1.15 eV
T ₁	0.08 eV	0.82 eV
³ MLCT	3.91 %	9.78 %



Target-in rate of candidates suggested by **human vs. ML**

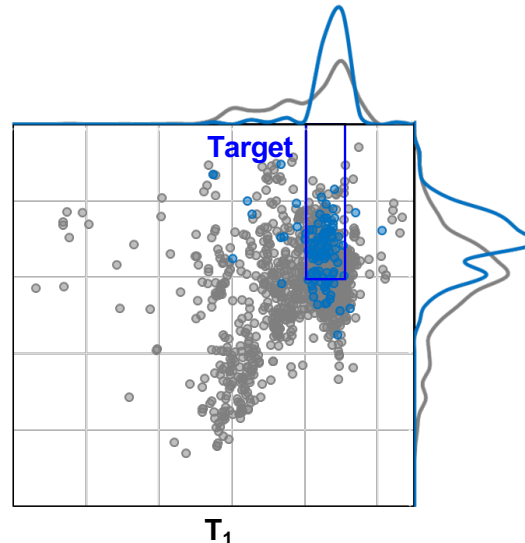
- **Pre-screening by a higher ML-predictability scale** further increases the ML-inference accuracy.
⇒ **Higher target-in rate** of virtual HTS by ML inference
→ **Higher efficiency** to find a material w/ the properties of interest in the target range

Screened by
intermediate predictability scale
and ML-inferred T_1 & $^3\text{MLCT}$

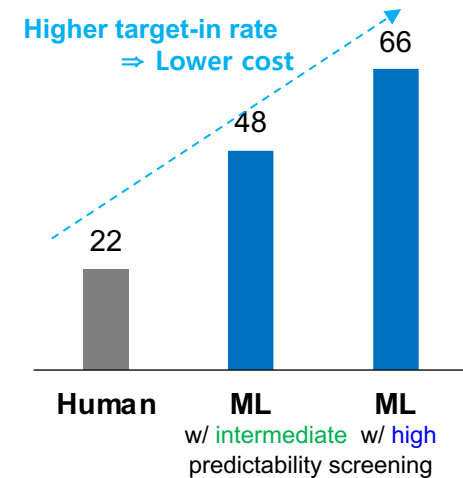


● DFT properties suggested by human (used for training ML models as well)
● DFT properties of target-in candidates by ML inference

Screened by
High predictability scale
and ML-inferred T_1 & $^3\text{MLCT}$

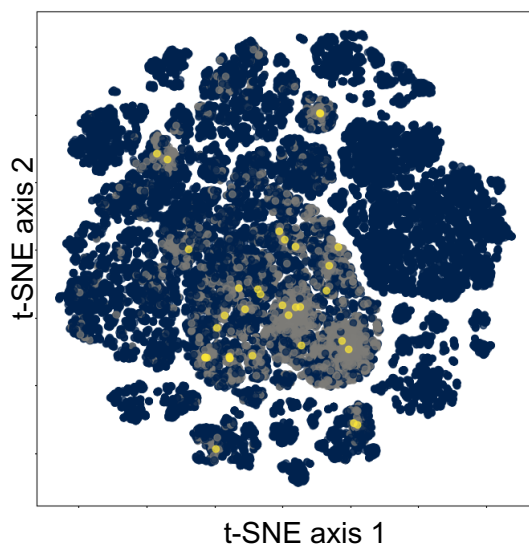


Target-in rate [%]
(Both for T_1 & $^3\text{MLCT}$)



Summary & future works

- The current work has successfully implemented a virtual HTS of materials based on a good performance on materials property predictions using machine learning models **just with 1000 data records**.
- Pre-screening based on a **ML-predictability** scale has allowed **≥700K molecules** to be examined as the virtual HTS candidate molecules.
- Future works will involve **a wise selection of molecules** to build a bigger database of DFT properties **to expand the coverage of ML models** for the chemical space of our interest.



- ML trained structures **1K** (0.07%)
- Structure DB w/ higher ML-predictability: **780K** (52%)
Current ML-prediction coverage in the chemical space of our interest
- Structure DB w/ lower ML-predictability: **720K** (48%)
ML-prediction coverage planned for the future

Summary & future works

